# Improving access to interchanges between material and immaterial cultural heritage through semantic modeling

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## Abstract

Semantic modeling can play an important role in enhancing accessibility to the immaterial culture related to artifacts. To this end, we examine whether the domain standards CIDOC-CRM and LRMoo can express the interactions of an artwork with the contexts it traverses through a case study of XVII Century integrative restoration.

## Introduction

Although UNESCO defines culture as a set of spiritual, material, intellectual, and emotional features of a society or social group, it distinguishes between tangible and intangible cultural heritage (CH). This distinction is not only problematic for preservation but also causes a loss of accessibility to holistic information and experiences of CH. Secondly, despite adaptations of the most diffused CH ontology, CIDOC-CRM (CRM) by modeling creative processes and preservation practices, it is still problematic to capture interactions between material and immaterial cultural expressions (Bannour/ Marinica 2018; Zlodi 2023: 115-148). This hinders the understanding of changes of cultural objects in context. This preliminary work examines whether available ontologies allow modeling interactions of artworks with the contexts they traverse. This will be illustrated by discussing the impact of changing ideas about the restoration of the Torlonia statue representing Isis on the evolution of its iconographic meaning.

A semantic model for this change will enrich artefact data descriptions in LAM centres, tracking the work's material aspects and the immaterial phenomena that have shaped the object's history.

## Case Study

Restoration practices enable exploring the different stages of an artwork's history by adding or removing parts shaped by society's taste. In the 17th century, Roman noble residences displayed ancient sculptures as symbols of lineage. Elite families employed artists to restore incomplete pieces, combining modern additions with ancient fragments (de Lachenal, 2020: 79-91). This approach was applied to the 'Isis Restored as Ceres,' now housed in Villa Torlonia[1] . The statue depicts a female figure identified as Isis, dressed in a fringed chiton. It exemplifies the fusion of ancient and modern elements: the original dark brown marble bust, dated between the 4th and 1st centuries BC, was supplemented with modern additions, including the neck, diadem, and forearms, as well as ancient ones for the head and feet (Capaldi 2020: 248-249).

The original Roman fragment depicting Isis, recognizable iconographically thanks to the knot in the robe, was modified during the 17th century to suit the latest taste of Roman collectors. This change involved a physical integration (adding missing limbs), and a conceptual transformation. The sculpture was restored as Ceres, the Roman goddess of fertility, identifiable by the attributes of ears of corn held in her right hand (Buccino 2002: 203-207). In short, restoration practices resulted in a double metamorphosis of the work, material and immaterial, through additions that transformed the subject.

## Modeling

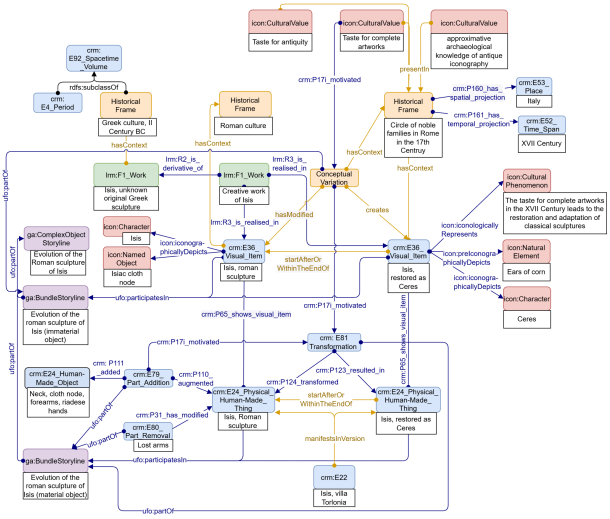
Based on our examination of the characteristics of the case study, we identified the following core modeling requirements (Req). The model should:

**Req1**: trace the evolution of immaterial aspects related to material modifications;

**Req2**: represent the different versions of the object and the actions that modify them;

**Req3**: represent the object as an individual entity despite its modifications and iconographic changes;

**Req4**: relate the object to its contemporary cultural context to enhance understanding of modifications, establishing connections between cultural traits and object changes.




*Model representing the case study requirements.*

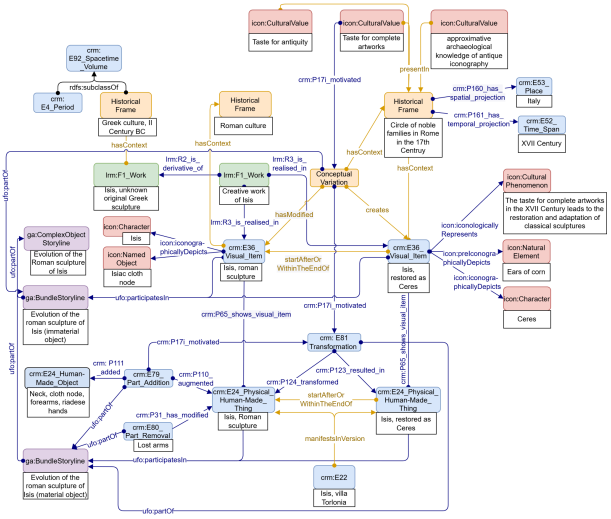
Following ontology development good practices (Carriero et al. 2020), we evaluate whether existing domain-related ontologies effectively express the features outlined in our requirements (Fig. 1), considering key standards from the museum and library communities: CRM (Bekiari et al. 2024a) and FRBR through the extension LRMoo (Bekiari et al. 2024b), aligning the two models. We also consider domain-specific models, such as ICON (Baroncini 2024; Sartini et al. 2023) for subjects and cultural phenomena and Storylines (van den Heuvel/Zamborlini 2021) for object evolution over time. Classes and properties that completely or partially fulfil the requirements, presented in Figs. 2-3, are discussed below.

**Req1**: Represent Object evolution as an ObjectStoryline, with separate narratives for material and immaterial changes.

**Req2**: Represent the different changes at conceptual and physical level.

FRBR describes a creative work according to four layers (i.e., Work, Expression, Manifestation, and Item), from the conceptual creation to the physical item, to represent the variations among editions in the library domain. In the last decades, several works applied the model to the arts (Bacca/Clarke 2007; Nakasone 2023). Nevertheless, the particular case of restorations changing the artwork's conceptual level was not fully formalized (Macaluso 2024).

In our case, a physical modification (crm:E79_Part_Addition) adds new attributes to the physical object (crm: E24_Human-Made_Object, see Fig. 2), creating a new Manifestation and Expression, as the subject changes from Isis to Ceres. Nevertheless, LRMoo does not include descriptions of physical modifications influencing immaterial classes. Furthermore, the physical variation cannot be expressed by a Manifestation, which has an abstract nature, unlike the class crm:E81 Transformation.



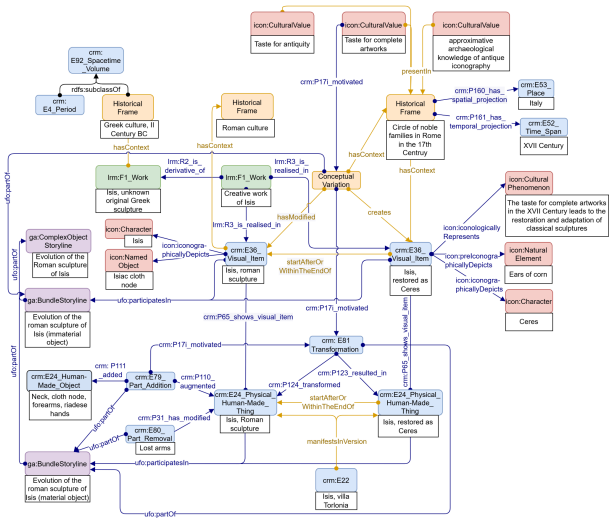

*Representation of physical modifications with CRM (Req2)*

**Req3**: Trace the artwork's identity throughout its evolution, as its substance persists despite changes in subject interpretation.

crm:E81_Transformation only indicates that physical changes result in identity changes, creating a new physical object (Grillo 2022).

**Req4**: Represent a broader cultural context shaped by heterogeneous phenomena.

Add a new class HistoricalFrame, subclass of crm:E92_Spacetime_Volume, that spatiotemporally encompasses cultural contexts including heterogeneous phenomena. The class icon:CulturalValue is suitable for expressing the cultural traits of a context.




*Modeling of the Isis sculpture with existing ontologies (proposed classes and properties in orange)*

# Discussion

From this preliminary study, it emerged that, despite some qualities of FRBR for capturing variations at multiple levels, its implementation for representing intellectual changes in the same work challenges its application to our scenario. CRM enables registering physical modification and transformation in objects, but its proposed classes do not fully allow tracing the relationships between the object's material changes and conceptual transformations over time. Whereas the concept of contextual culture can be partially aligned to the crm:E4_Period class, and ICON's classes can describe cultural traits, further classes and properties are needed to better describe the relation between the context and the object versions.

# Conclusions

This work revealed that current CH standards only partially enable capturing the evolution of cultural artifacts in their cultural contexts, and insufficiently address interactions between material and immaterial CH. Using a case study on integrative restoration, key requirements were formulated for ontological modeling and assessed to which extent existing domain standards (CRM, LRMoo) could cover these. The results highlight a need to extend existing ontologies with other typologies of material and immaterial culture interactions in the semantic web realm.

# Notes

1. https://www.fondazionetorlonia.org/statue-isis-restored-ceres

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